Chapter 8: Instruction Encoding and ISA

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N4-02c-92

Learning Objectives

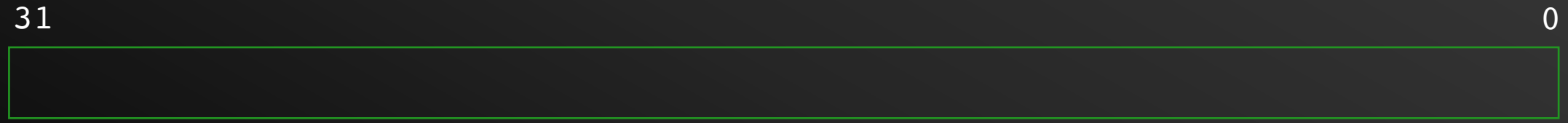
- Describe the instruction format of ARM instruction-set architecture (ISA)

- Describe differences between fixed and variable-length ISA

Instruction Encoding

- For CPU to identify each unique assembly instruction, they must be encoded with unique binary patterns.
  - ARM 32-bit machine encode instructions in a 32-bit word

- What is included?
  - Instruction type ➔ encoded
  - Operand(s) ➔ encoded
  - Condition for conditional execution ➔ encoded
  - Other flags?

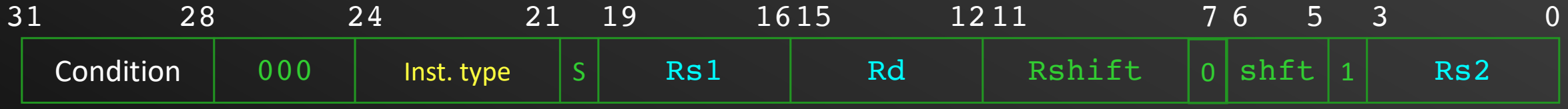


Overall Instruction Format

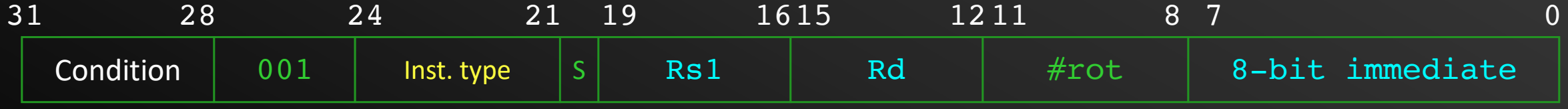
- Arithmetic & Logic instructions
  - ADD, ADC, SUB, SBC, RSB, AND, EOR, RSC, ORR, BIC

XXXCCS Rd, Rs1, Rs2 , {shft #}

XXXCCS Rd, Rs1, Rs2 , {shft Rshift}



XXXCCS Rd, Rs1, #immediate (rotated)

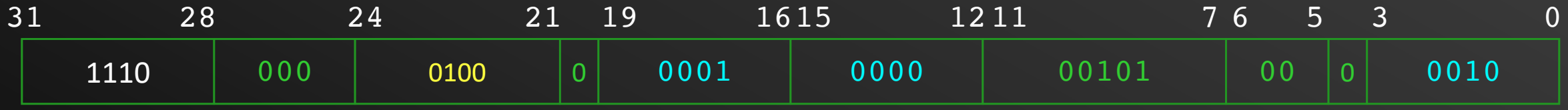


Overall Instruction Format

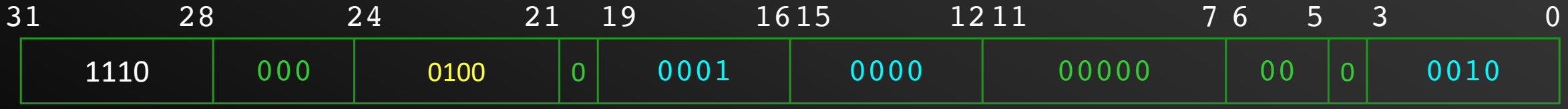
- Arithmetic & Logic instructions
  - ADD, ADC, SUB, SBC, RSB, AND, EOR, RSC, ORR, BIC

XXXCCS Rd, Rs1, Rs2 , {shft #}

ADD R0,R1,R2,LSL #5



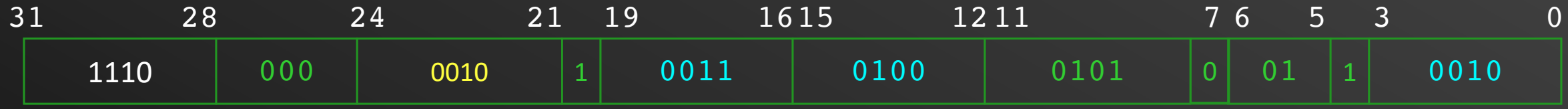
ADD R0,R1,R2



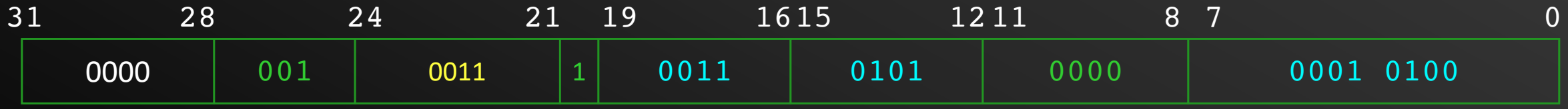
Overall Instruction Format

- Arithmetic & Logic instructions
  - `ADD, ADC, SUB, SBC, RSB, AND, EOR, RSC, ORR, BIC`

`SUBS R4, R3, R2, LSR R5`



`RSBEQS R5,R3,#20`

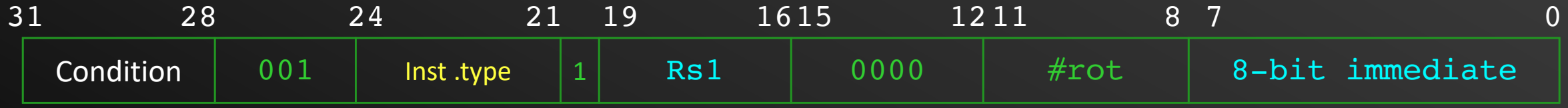


Overall instruction format

- Comparison instructions
  - `TST, TEQ, CMP, CMN`

`XXXCC Rs1, Rs2 , {shft #}`

`XXXCC Rs1, #immediate (rotated)`

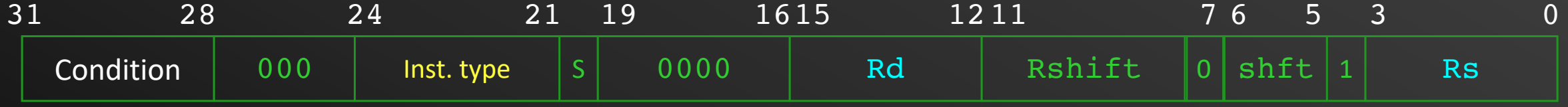


Overall instruction format

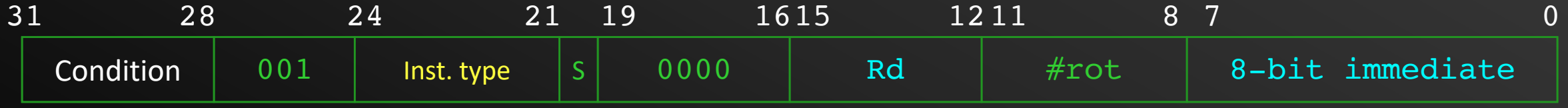
- MOV instructions
  - MOV, MOVN

XXXCCS Rd, Rs , {shft #}

XXXCCS Rd, Rs , {shft #}



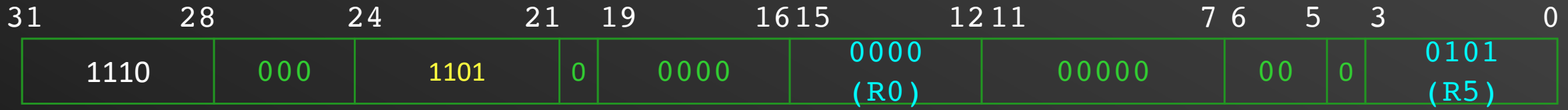
XXXCC Rd, #immediate (rotated)



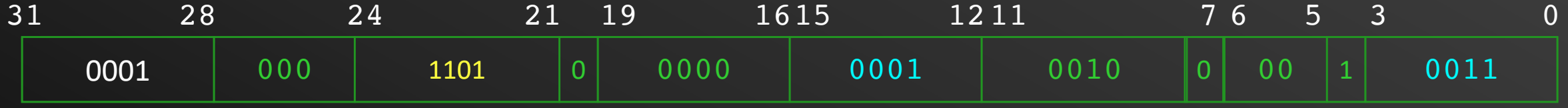
Overall instruction format

- MOV instructions
  - MOV, MOVN

MOV R0,R5



MOVNE R1, R3, LSL R2



MOV R1,#0x40000000

Inst. Type field Field

- 4 bits for instructions ➔ up to 16 combinations
- 4 bits for source and destination registers ➔ 16 registers

Code	Instruction
0000	AND
0001	EOR
0010	SUB
0011	RSB
0100	ADD
0101	ADC
0110	SBC
0111	RSC

Code	Instruction
1000	TST
1001	TEQ
1010	CMP
1011	CMN
1100	ORR
1101	MOV
1110	BIC
1111	MVN

Condition Field

- 4 bits for condition → up to 16 difference combinations

Code	Condition	Flags	Meaning
0000	EQ	Z = 1	Equal
0001	NE	Z = 0	Not equal
0010	CS or HS	C = 1	Higher or same, unsigned
0011	CC or LO	C = 0	Lower, unsigned
0100	MI	N = 1	Negative
0101	PL	N = 0	Positive or zero
0110	VS	V = 1	Overflow
0111	VC	V = 0	No overflow
1000	HI	C = 1 and Z = 0	Higher, unsigned
1001	LS	C = 0 or Z = 1	Lower or same, unsigned
1010	GE	N = V	Greater than or equal, signed
1011	LT	N != V	Less than, signed
1100	GT	Z = 0 and N = V	Greater than, signed
1101	LE	Z = 1 and N != V	Less than or equal, signed
1110	AL		Always.
1111	NV		Reserved (unused)

What about shift instructions, what's there instruction format?

LSL, LSR, ASR, ROR, and RRX has no dedicated instructions

Shift/Rotate instructions

- Instructions are encoded as MOV instructions

  E.g. LSL R1, R2, #5 ≡ MOV R1, R2, LSL #5

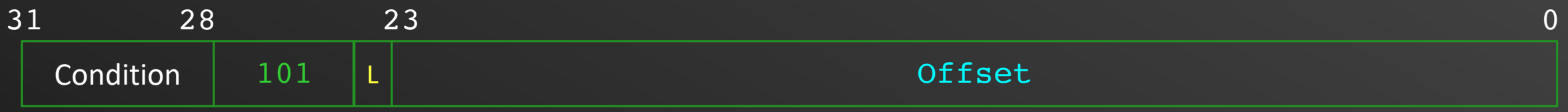
- Specify the shift type in bits 5 and 6

Code	Shift type
00	LSL
01	LSR
10	ASR
11	ROR/RRX

Branch Instructions

- For conditional branch and branch with link

```
B(L)CC   Offset
```



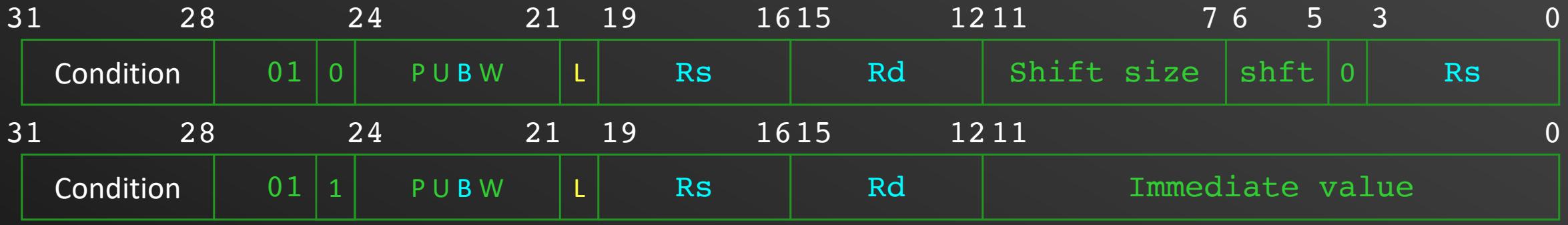
- 26 bits offset shifted to the right by 2 (branching to word addresses) ➔ calculated by the assembler

- Branch range: ±32 MBytes

Load/Store Instructions

- Load/Store word bye

```
LDR/STR{B}CC   Rs,[RD,offset pre or post]W
```

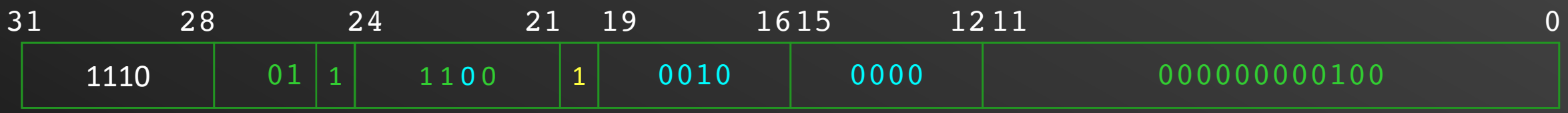


- L bit determines load (1) or store (0)
- P bit determines indexing pre (1) or post (0)
- U determines offset direction add (1) or subtract (0)offset
- B determines byte (1) or word (0) access
- W is for write back (!) of the offset

Load/Store Instructions

- Load/Store word bye

```
LDR   R0,[R2,#4]
```



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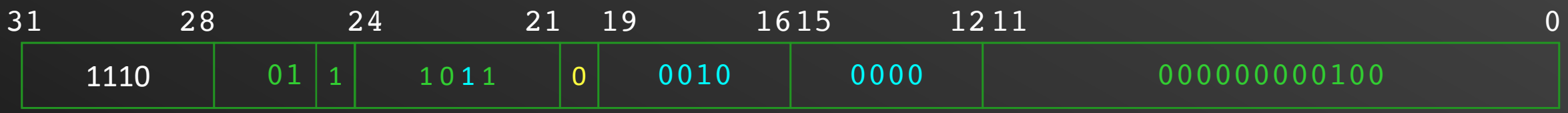
Load/Store Instructions

- Load/Store word bye

```
STRB R0,[R2,#-4]!
```



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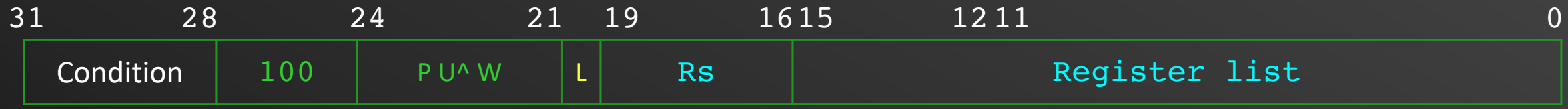
Load/Store Instructions

- Load/Store Multiple instructions

```
LDM/STMFXCC   RsW,{register list}
```



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- ^ is “don’t care”
- W is for write back (!) after operation

Load/Store Instructions

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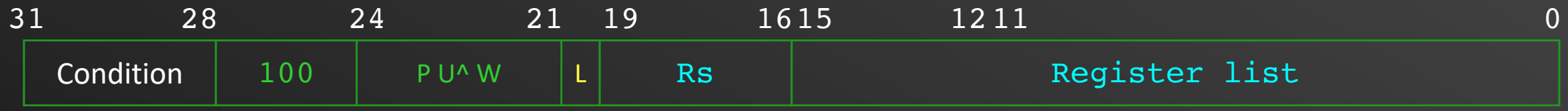

- L bit determines load (1) or store (0)
- P bit determines indexing before (1) or after ((
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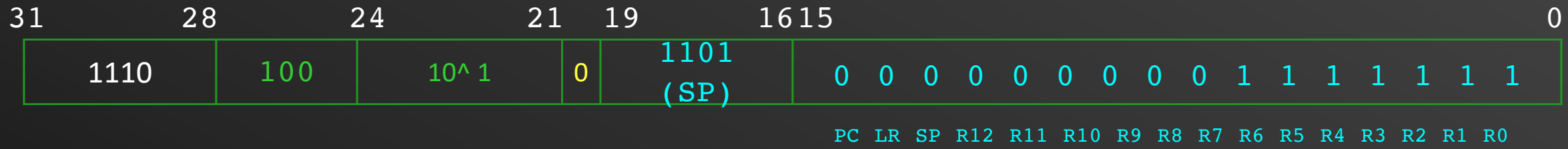
PU	Instruction
10	STMFD
01	LDMFD
00	STMFA
11	LDMFA

Load/Store Instructions

- Load/Store Multiple instructions

```
STMFD SP!,{R0-R6}
```



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Fixed vs. Variable-Length instruction

- ARM is fixed-length
  - Operands had to be registers or immediate values

- Variable length instructions
  - Intel 80x86: Instructions vary from 1 to 17 bytes long.
  - Digital VAX: Instructions vary from 1 to 54 bytes long.
  - Require multi-step fetch and decode.
  - Allow for a more flexible (but complex) and compact instruction set.

Variable-Length: Intel Instruction Format

- Complex instructions = complex hardware
- Higher variety for smaller code and faster execution

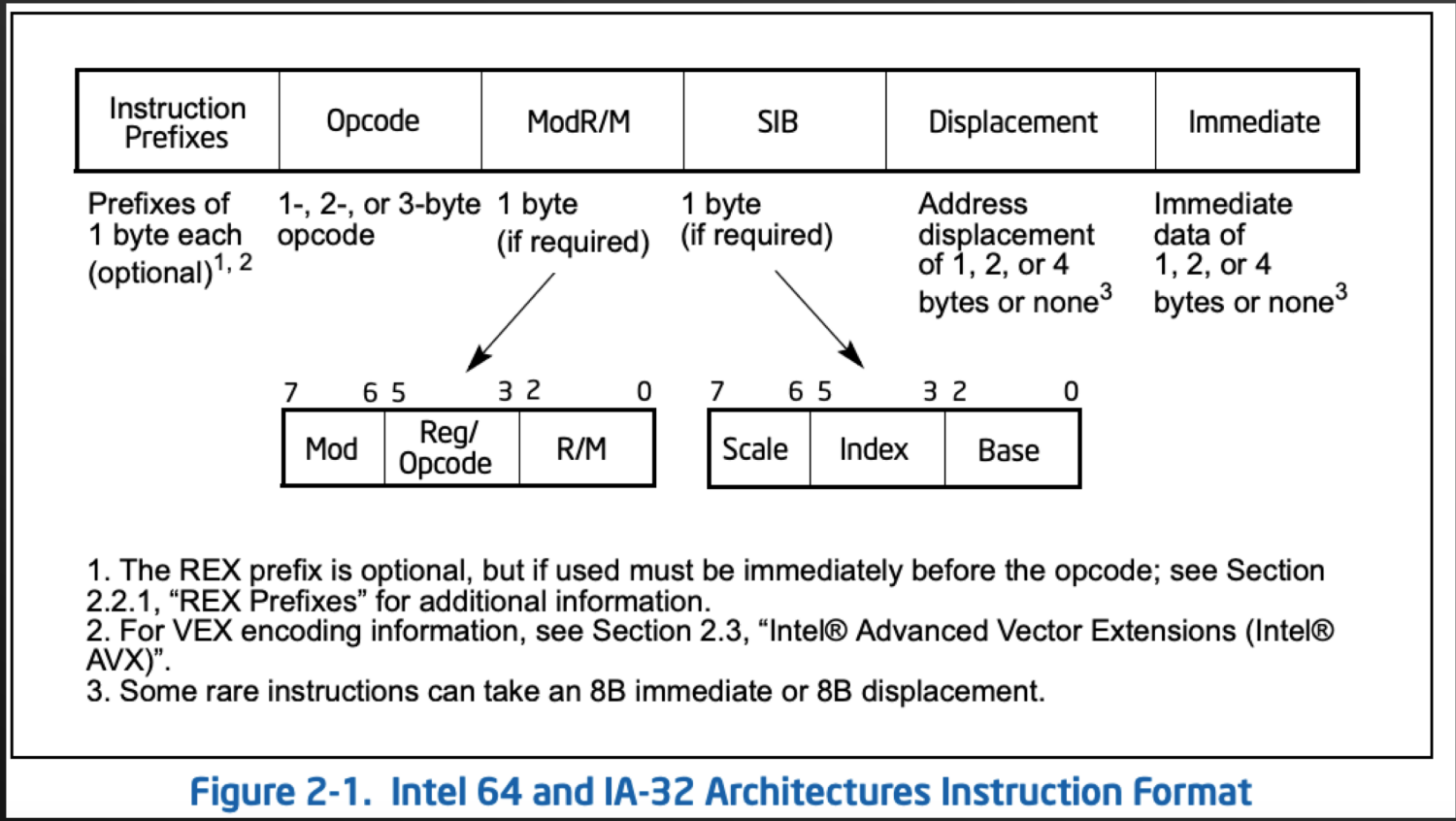
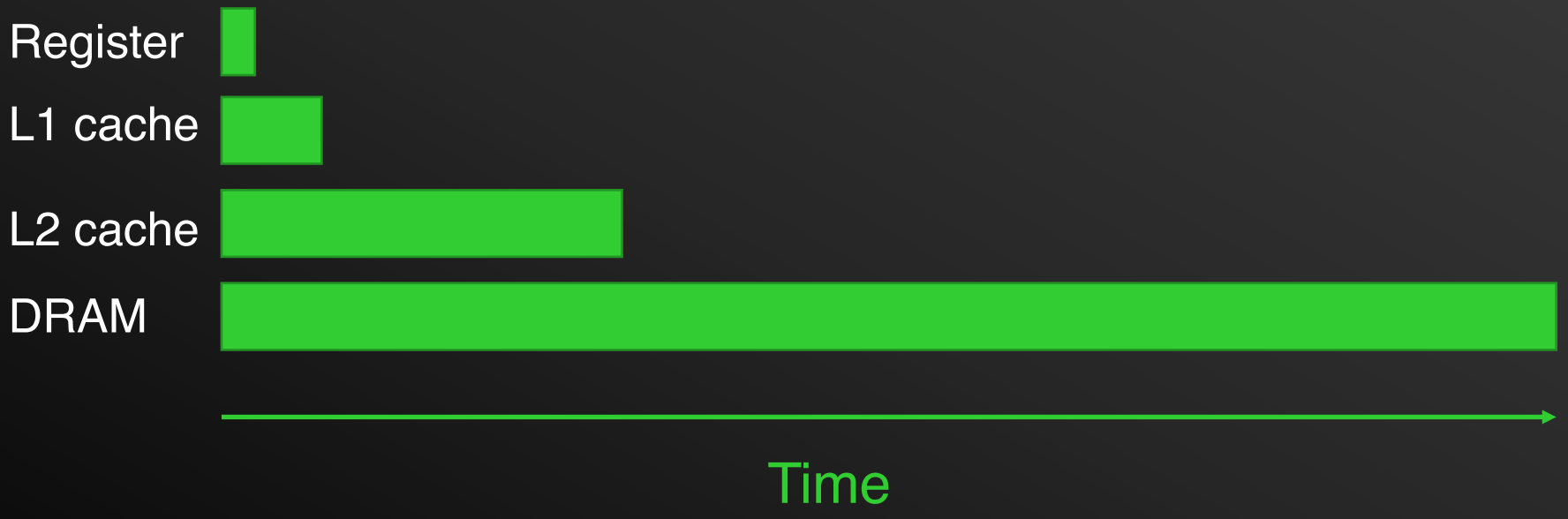


Figure 2-1. Intel 64 and IA-32 Architectures Instruction Format

Using Registers

- Operations using registers are **faster** than those involving memory.
  - Data transfer between registers and ALU is faster due to it physically proximity.
  - To speed up code execution, keep as much of your computations within **registers**.



Implication of Register Count

- Implementing many registers within the CPU will incurs increasing overheads.
  - They take up precious **space** in the silicon die.
  - Connections to various busses increase the **routing** and **multiplexing complexity**.
  - More registers increases the **operand size** during instruction encoding.

ADD (3)

15	14	13	12	11	10	9	8 – 6	5 – 3	2 – 0
0	0	0	1	1	0	0	Rm	Rn	Rd

This form of ADD adds the value of one register to the value of a second register, and stores the result in a third register. The condition code flags are updated, based on the result.

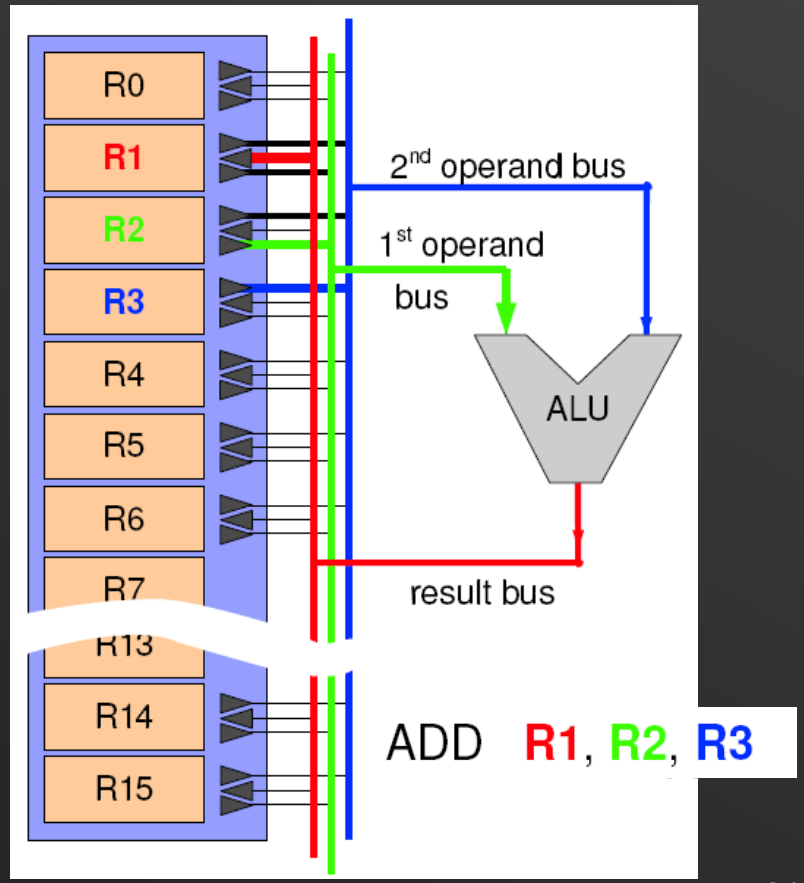
Syntax

`ADD <Rd>, <Rn>, <Rm>`

where:

`<Rd>`	Is the destination register for the completed operation.
`<Rn>`	Specifies the register containing the first value for the addition.
`<Rm>`	Specifies the register containing the second value for the addition.

Instruction Encoding for ADD in Thumb-2



Orthogonality of ISA

- In a truly orthogonal ISA, every instruction is able to use every available addressing modes.
- A truly orthogonal ISA does not restrict certain instruction from using only specific registers.
- Difficult to achieve complete orthogonality due to the **large number of bit patterns** required to express all combinations of operations and addressing modes.
- Increase instruction length will increases the memory size required by the program.

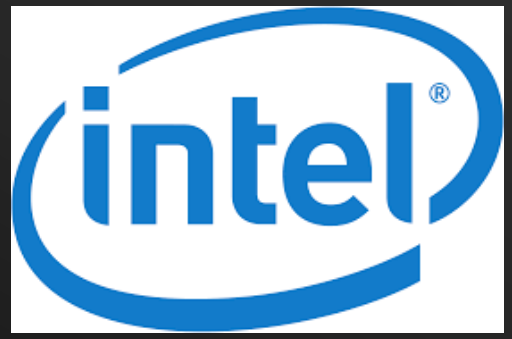
Summary - Why is Good Instruction Set Architecture (ISA) Design Important?

- Good ISA design can yield benefits such as:

  - Increases the execution performance of the processor.
  - Increases the code density (i.e. how much memory a piece of code will occupy).
  - Reduces the power consumption of the CPU.
  - Reduces manufacturing cost of processor.
  - Easy for programmers to write efficient programs.
  - Efficient mapping of high-level programming requirements into low-level instruction sets.

Why Should I Care About Processor Designs and ISA?

- Designing a processor (or a co-processor) is no longer a hardware company job



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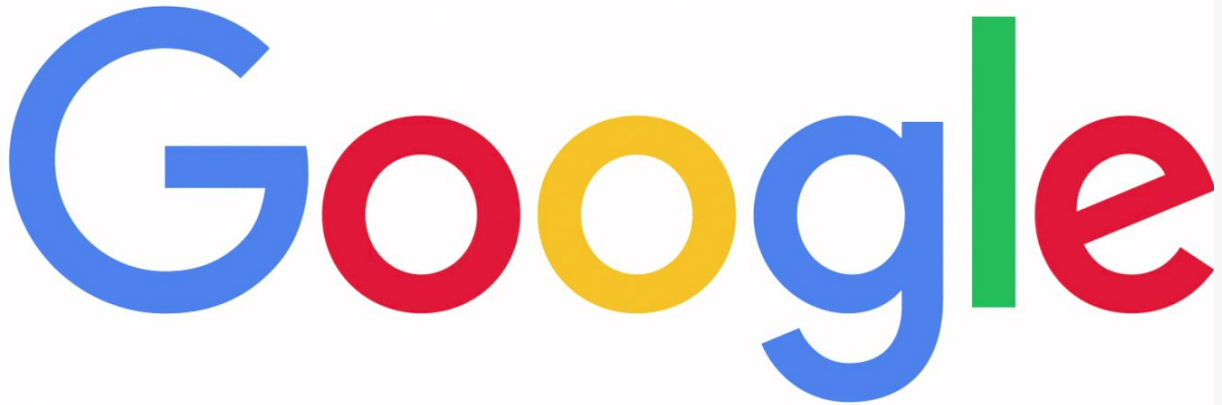
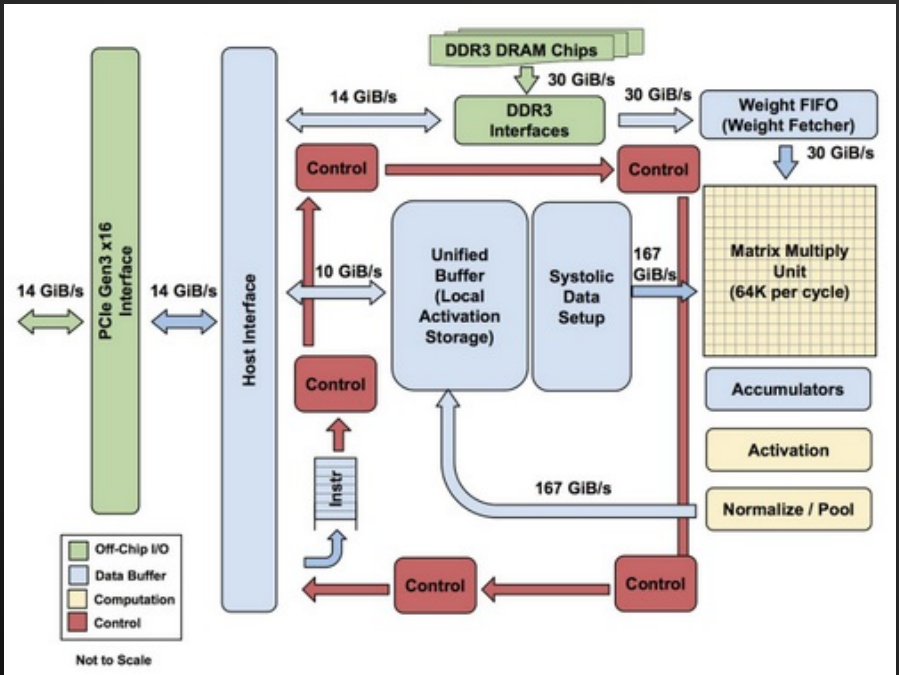
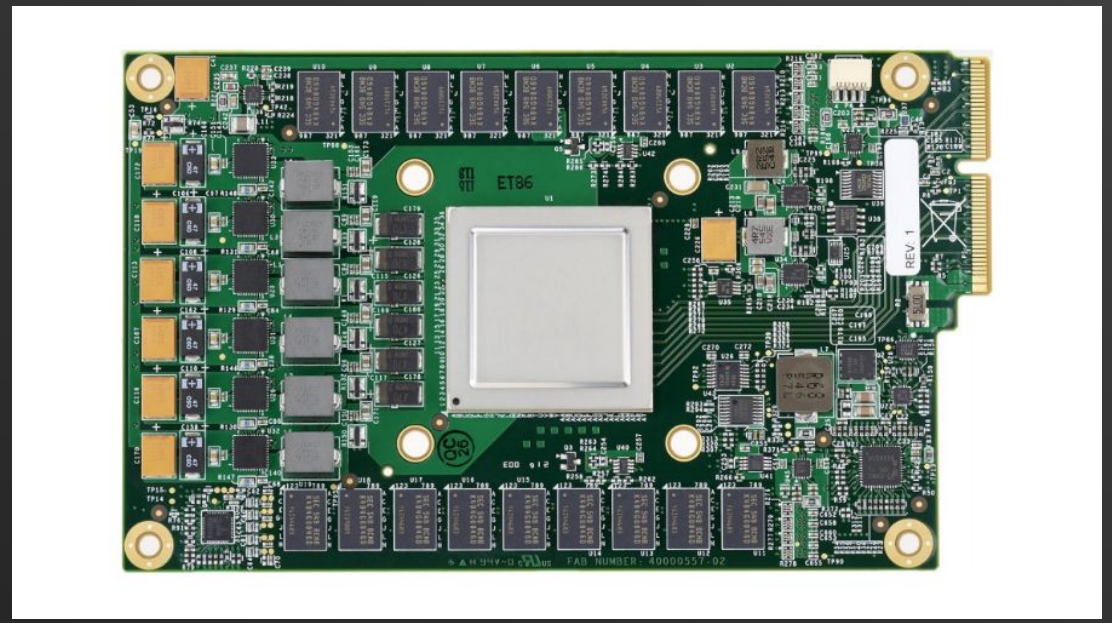


The Google TPU

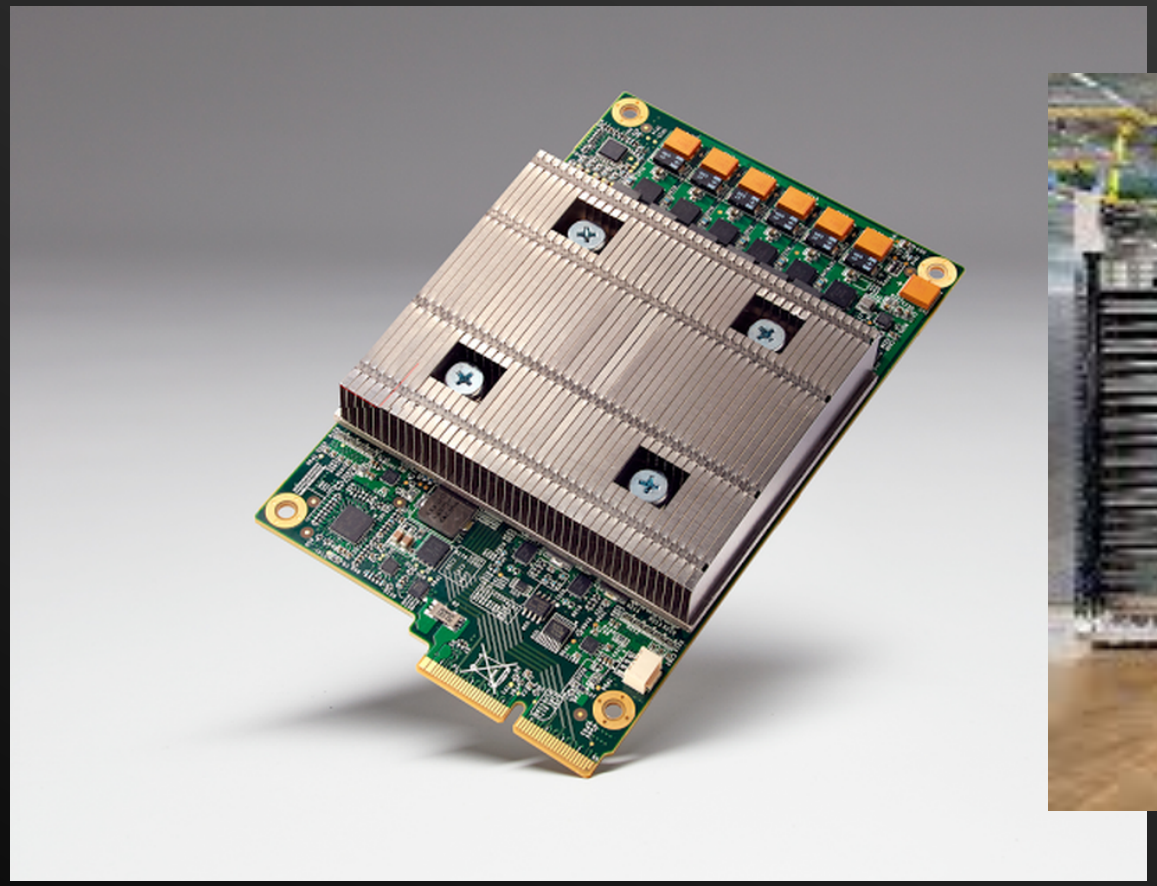
- Tensor Processing Unit
- Designed by Engineers in Google
- Tailored to make AI applications run faster





The Google TPU

- Design by Engineers in Google to make AI applications run faster



Summary

- ARM is fixed-length load/store architecture
  - Operands reside in registers
- Other architectures can use variable-length format
  - E.g, Intel architecture
- Use of registers is key towards higher performance